labsession1

2026-07-06

R Setup

Data import

Data comes from HMD and HFD dataset for Spain in 2014.

```
b <- read_csv("https://raw.githubusercontent.com/timriffe/BSSD2026Module2/refs/heads/master/data/ES_B20
d <- read_csv("https://raw.githubusercontent.com/timriffe/BSSD2026Module2/refs/heads/master/data/ES_D20
p <- read_csv("https://raw.githubusercontent.com/timriffe/BSSD2026Module2/refs/heads/master/data/ES_P20
```

Tasks

task 1: calculate exposures

Let's calculate the exposure based on averaging the population from the start and end of the year. Here, we are going to use this strategy.

$$E = \frac{P(1) + P(2)}{2}$$

We are going to work with tidy dataset: there is a (or some) key variables and the observation.

```
e <- p |>
  select(-open_interval) |>
  pivot_longer(
    female:total,
    names_to = "sex",
    values_to = "pop"
  ) |>
  # inventing column pop 1 and pop2 and taking the average based on that
  # this will work for long time series and we will do that based on lead() and lag()
  arrange(age, sex, year) |>
  mutate(
    pop2 = lead(pop),
    .by = c(age, sex)
  ) |>
  filter(!is.na(pop2)) |>
  # computing exposure
  reframe(
    pop_exp = (pop + pop2) / 2,
    .by = c(age,sex)
  )
```

task 2: calculate the fertility rates

```r
e |>
  filter(sex == "total") |>
  left_join(
    b |> select(-year, -open_interval),
    by = join_by(age)
  ) |>
  mutate(
    total = if_else(is.na(total),0,total)
  ) |>
  summarise(
    b = sum(total),
    e = sum(pop_exp)
  ) |>
  mutate(
    CBR = b / e * 1000
  )
```

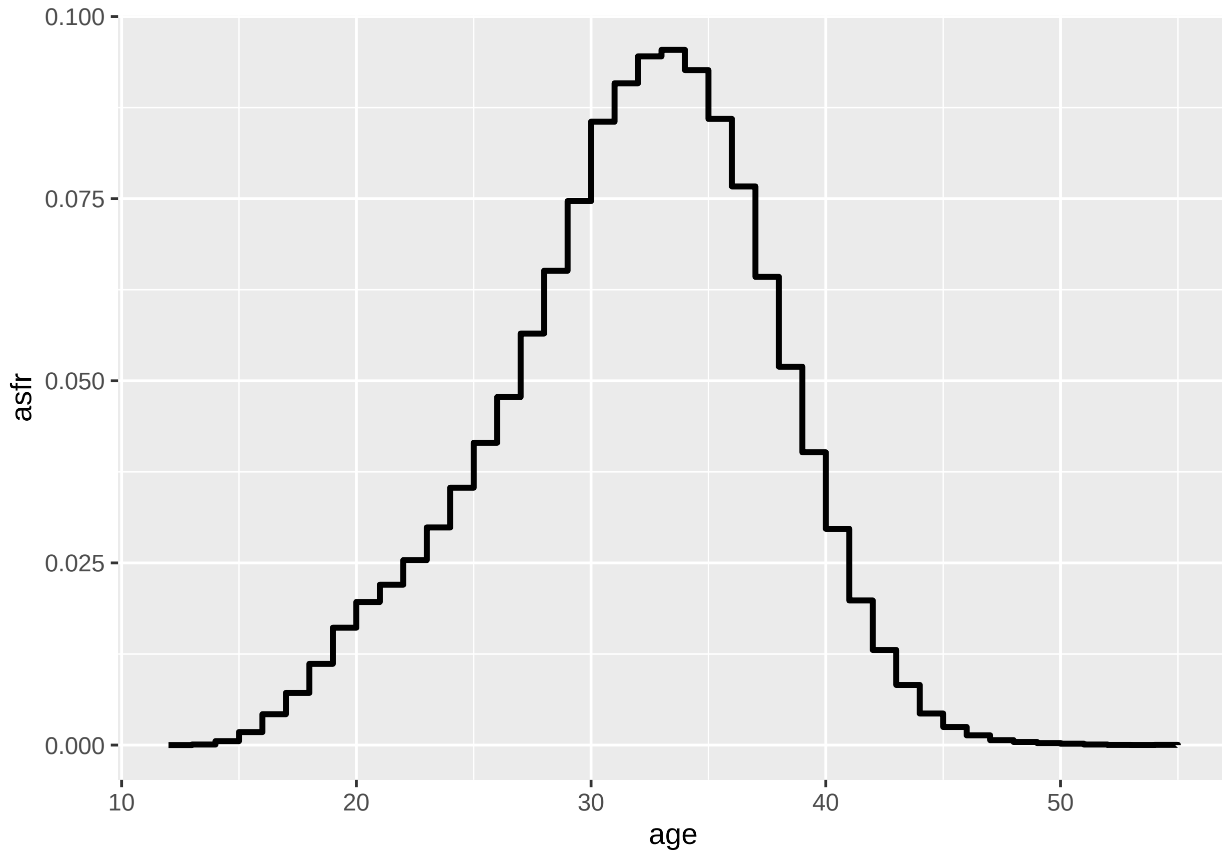
```
## # A tibble: 1 x 3
##        b         e   CBR
##    <dbl>     <dbl> <dbl>
## 1 426076 46443889.  9.17
```

exercise: age-specific fertility rates

Join to female, and calculate the rate doing by age.

```r
asfr <- e |>
  filter(sex == "female") |>
  right_join(
    b |> select(-year, -open_interval),
    by = join_by(age)
  ) |>
  mutate(
    asfr = total / pop_exp
  )

# plotting it
asfr |>
  ggplot() +
  aes(x = age, y = asfr) +
  geom_step(linewidth = 1.05)
```



We can add up the ASFR to get the TFR:

```
asfr |>
  summarise(tfr = sum(asfr))
```

```
## # A tibble: 1 x 1
##     tfr
##   <dbl>
## 1  1.32
```

This is a synthetic cohort, and we assume that they all survived untill the end of the reproductive period.

mortality rates

exercise: calculate age-specific mortality rates

Let's do similarly for fertility

```
asmr <- d |>
  select(-year, -open_interval) |>
  pivot_longer(
    c(female, male, total),
    names_to = "sex",
    values_to = "D"
  ) |>
  left_join(
    e,
    by = join_by(sex, age)
  ) |>
```

```
  # computing asmr
  mutate(asmr = D / pop_exp)

# plotting it

asmr |>
  filter(age <= 109) |>
  ggplot() +
  aes(x = age, y = asmr, color = sex) +
  geom_line(linewidth = 1.05) +
  scale_y_log10()
```